比特率与波特率

比特率(bit rate)：R_b (bit/sec) 对应的 $T_b=\frac{1}{R_b}$ (传输一个bit所需的时间)

波特率(baud rate)：R_s (sys/sec 或 baud) 对应的 $T_s=\frac{1}{R_s}$ (传输一个symbol所需的时间)

带宽(Bandwidth)：W (or B) Hz

抽样频率(Sampling frequency)：f_s Hz 有 $f_s\geq 2f_m$ (f_m是模拟信号的最大频率)

Nyquist capacity (理想低通，无码间串扰) $R_s\leq C_{baud}=2W$

Shannon capacity (非理想，有限带宽，高斯噪声) $R_b\leq C_{bits}=W\log_2(1+SNR)$

主瓣带宽=第一零点带宽$=W=\frac{1}{\tau}$，其中τ是有效时间

采用滚降(Rolloff) $R_s=\frac{2B}{1+r}$ ($0\leq r\leq 1$，B为系统的绝对带宽，r为滚降系数)

量化(Quantization)时，进行L阶量化：$L=2^l$ $R_s=lf_s$

M位bit表示时：$M=2^n$ $R_b=nR_s$ 则有 $R_b=nlf_s$

频带利用率(Spectral efficiency)：$\eta=\frac{R_b}{B}$ (bps/Hz)

误码率：$P_e=\frac{N_e}{N}$ (是不是已经删掉了呀？)(应该是)

信噪比SNR(Signal to Noise Ratio)：$SNR=\frac{\text{信号功率}}{\text{噪声功率}}=\frac{P}{P_n}=\frac{E_bR_b}{N_0W}=\eta\frac{E_b}{N_0}$ (E_b是传输一位bit的平均能量)

信息量与信息熵

信息量：$I=\log_2\frac{1}{p}$

信息熵(Information entropy)：$H=E(I)=\sum_{i=1}^{N}P_i\log_2\frac{1}{P_i}$ (当$P_i=\frac{1}{N}$时，有最大信息熵$\log_2N$)

条件信息熵：$H(X|Y)=-\sum_{x,y}P(x,y)\log_2P(x|y)$

互信息：$I(X;Y)=H(X)-H(X|Y)=H(Y)-H(Y|X)$

信源编码

平均编码长度：$L=E(n)=\sum_{i=1}^{N}P_in_i$

压缩率：$r=\frac{\text{无编码长度}}{\text{平均编码长度}}$

编码效率：$\eta=\frac{H}{L}$

信道编码

编码率：$R_c=\frac{k}{n}$ $r=n-k$

最小汉明距离：d_{min}

检错能力：$e\leq d_{min}-1$

纠错能力：$t\leq\frac{d_{min}-1}{2}$

生成矩阵(Generated matrix)：$G=[I_k|P]$

监督矩阵(Parity check matrix)：$H=[P^T|I_{n-k}]$

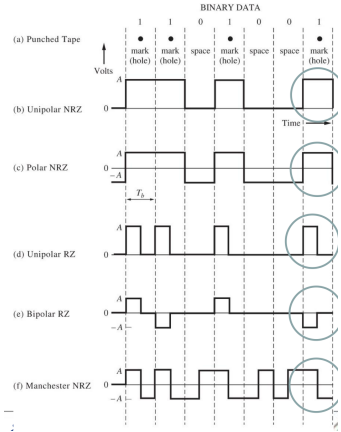
调制

脉冲编码调制 (Pulse Code Modulation)

量化信噪比 $SQNR=\frac{S_q}{N_q}=\frac{(M^2-1)q^2}{q^2}=M^2-1$

M阶量化时 $M=2^n$ $SQNR=(4.8+6n-\alpha)$ dB

脉冲幅度调制 (Pulse Amplitude Modulation)



Code Type	First Null Bandwidth (Hz)	Spectral Efficiency $\eta=R/B$ [(bits/s)/Hz]
Unipolar NRZ	R	1
Polar NRZ	R	1
Unipolar RZ	$2R$	$\frac{1}{2}$
Bipolar RZ	R	1
Manchester NRZ	$2R$	$\frac{1}{2}$
Multilevel polar NRZ	R/ℓ	ℓ

功率谱密度(Power Spectrum Density) $P(f)=\frac{|F(f)|^2}{T_s}\sum_{k=-\infty}^{\infty}R(k)e^{-2\pi kfT_s}$ 其中 $F(f)$ 是脉冲 $f(t)$ 的傅里叶变换，$R(k)$ 是自相关系数

Unipolar NRZ:

$R(0)=\sum_{i=1}^{2}(a_ia_{i+0})P_i=A^2\times\frac{1}{2}+0^2\times\frac{1}{2}=\frac{1}{2}A^2$ $R(k)=\sum_{i=1}^{2}(a_ia_{i+k})P_i=A^2\times\frac{1}{4}+A\cdot0\times\frac{1}{4}+0\cdot A\times\frac{1}{4}+0^2\times\frac{1}{4}=\frac{1}{4}A^2$

$R_{unipolar}(k)=\begin{cases}\frac{1}{2}A^2 & k=0\\ \frac{1}{4}A^2 & k\neq0\end{cases}$ $f(t)=Rect(\frac{t}{T_b})\leftrightarrow F(f)=T_b\frac{\sin\pi fT_b}{\pi fT_b}$

$P_{unipolar\ NRZ}(f)=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2\left[1+\sum_{k=-\infty}^{\infty}e^{2\pi kfT_b}\right]=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2\left[1+\frac{1}{T_b}\sum_{k=-\infty}^{\infty}\delta(f-\frac{n}{T_b})\right]=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2\left[1+\frac{1}{T_b}\sum_{n=-\infty}^{\infty}\delta(f)\right]$

Polar NRZ:

$R_{polar}(k)=\begin{cases}A^2 & k=0\\ 0 & k\neq0\end{cases}$ $P_{polar\ NRZ}(f)=A^2T_b\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2$

Unipolar RZ: $F(f)=\frac{T_b}{2}\frac{\sin\pi fT_b/2}{\pi fT_b/2}$

$R_{unipolar}(k)=\begin{cases}\frac{1}{2}A^2 & k=0\\ \frac{1}{4}A^2 & k\neq0\end{cases}$ $P_{unipolar\ RZ}(f)=\frac{A^2T_b}{16}\left(\frac{\sin\pi fT_b/2}{\pi fT_b/2}\right)^2\left[1+\frac{1}{T_b}\sum_{n=-\infty}^{\infty}\delta(f-\frac{n}{T_b})\right]$

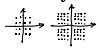
Bipolar RZ: $R(0)=\frac{1}{2}A^2$ $R(1)=-\frac{1}{4}A^2$ $R(k>1)=\sum_{i=1}^{2}(a_ia_{i+k})P_i=\frac{1}{8}A^2-\frac{1}{8}A^2=0$

$R_{bipolar}(k)=\begin{cases}\frac{1}{2} & k=0\\ -\frac{1}{4} & |k|=1\\ 0 & |k|>1\end{cases}$ $P_{bipolar\ RZ}(f)=\frac{A^2T_b}{8}\left(\frac{\sin\pi fT_b/2}{\pi fT_b/2}\right)^2[1-\cos(2\pi fT_b)]=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b/2}{\pi fT_b/2}\right)^2[\sin^2(\pi fT_b)]$

Manchester NRZ:

$P_{manchester\ NRZ}(f)=A^2T_b\left(\frac{\sin\pi fT_b/2}{\pi fT_b/2}\right)^2[\sin^2(\pi fT_b/2)]$

振幅键控(Amplitude-Shift Key) $S(t)=\begin{cases}A\cos(2\pi f_ct) & \text{binary 1}\\ 0 & \text{binary 0}\end{cases}$

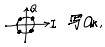
Quadrature Amplitude Modulation 

移频键控(Frequency-Shift Key) $S(t)=\begin{cases}A\cos(2\pi f_1t) & \text{binary 1}\\ A\cos(2\pi f_2t) & \text{binary 0}\end{cases}$

Multiple Frequency-Shift Key $S_i(t)=A\cos2\pi f_it$ $1\leqslant i\leqslant M$ $M=2^L$

$T_s=LT_b$ $2f_d=\frac{1}{T_s}$ 带需要带宽 $W_d=2Mf_d=\frac{M}{T_s}=\frac{2^L}{LT_b}$

移相键控(Phase-Shift Key) $S(t)=\begin{cases}A\cos(2\pi f_ct) & \text{binary 1}\\ A\cos(2\pi f_ct+\pi) & \text{binary 0}\end{cases}$

Quadrature Phase-Shift Key [多bit] 画 Constellation map  即 $a_k,b_k=\pm\frac{1}{2}$

扩频(Spread Spectrum)

跳频扩频(Frequency Hoping Spread Spectrum) using MFSK Fast $T_c<T_s$ /Slow $T_c\geqslant T_s$

直接扩频(Direct Sequence Spread Spectrum) using BPSK $S(t)=A\,d(t)\,c(t)\cos(2\pi f_ct)$

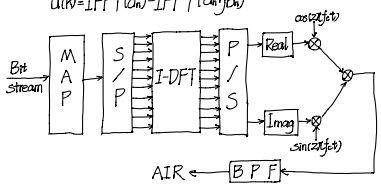
码分复用(Code-Division Multiplexing)/码分多址(Code-Division Multiple Access)

时分复用(Time-Division Multiplexing)/时分多址(Time-Division Multiple Access)

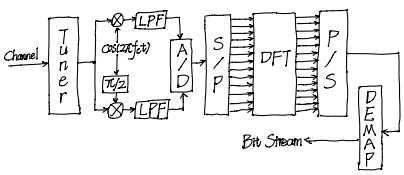
频分复用(Frequency-Division Multiplexing)/频分多址(Frequency-Division Multiple Access)

正交频分复用(Orthogonal Frequency Division Multiplexing)/正交频分多址(Orthogonal Frequence Division Multiple Access)

$u(t)=IFFT(d_n)=IFFT(a_n+jb_n)$



$d_n=FFT(u(k))$



信道传播

天线增益(Antenna Gain): $G=\frac{4\pi A_e}{\lambda^2}=\frac{4\pi f^2A_e}{c^2}$ A_e: 有效面积 λ: 波长 f: 波频率 c: 光速

空间损耗(Free Space Loss): $\frac{P_t}{P_r}=\frac{(4\pi d)^2}{\lambda^2}=\frac{(4\pi fd)^2}{c^2}$ P_t: 传输端信号功率 P_r: 接收端信号功率 d: 传输距离

热噪声(Thermal Noise)

单位热噪声功率: $N_0=KT\ (W/Hz)$ K=Boltzmann's constant $=1.3803\times10^{-23}J/K=-228.6\,dB\,J/K$ T: 开尔文温度

单位比特热噪声信噪比: $\frac{E_b}{N_0}=\frac{S/R}{KT}=\frac{S}{KTR}$

Diversity: Different phase relations will exist between the multipath rays from each antenna - so the interference will be different

频率复用: $a=ir\sqrt{3}$ $b=jr\sqrt{3}$ 

$N=i^2+j^2+ij$

$R_u=r\sqrt{3N}$ $r_c=r\sqrt{N}$